



Module 02 – Extra Class

BASIC STATISTIC

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Objectives

Introduction

- ❖ Random Variables
- ❖ Probability Distribution
- ❖ Mathematical Expectations (Mean, Median, Variance)

Application

- ❖ Covariance & Correlation

SECTION 1

Discrete Random Variables

SECTION 2

Mathematical Expectations

SECTION 3

Covariance & Correlation



Discrete Random Variables



A Random Variable

A random variable X is a function $X: \Omega \rightarrow \mathbb{R}$, maps an outcome $s \in \Omega$ to a number on the real line $X(s) \in \mathbb{R}$

A continuous random variable X is a function $X(s): \Omega \rightarrow \mathbb{R}$, maps an outcome s from **an uncountably infinite** to a number on the real line $X(s) \in \mathbb{R}$

A discrete random variable X is a function $X(s): \Omega \rightarrow \mathbb{R}$, maps an outcome from a finite or countably infinite sample space to a number on the real line $X(s) \in \mathbb{R}$

Discrete Random Variables



A Continuous Random Variable

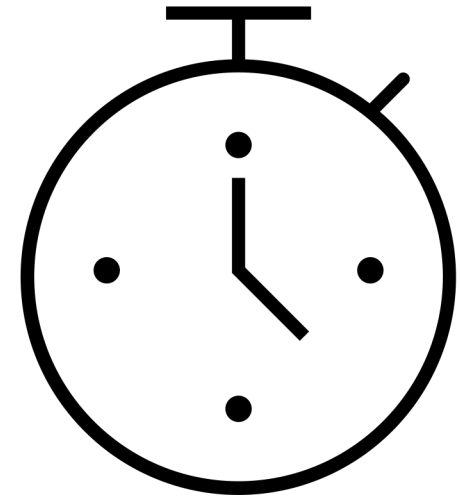
A continuous random variable X is a function $X(s): \Omega \rightarrow \mathbb{R}$, maps an outcome s from **an uncountably infinite** to a number on the real line $X(s) \in \mathbb{R}$

Rotate a pointer about a pivot in a plane (a clock)

Outcome: the angle where stops: $2\pi\theta, \theta \in (0,1]$

$$\Omega = (0, 1]$$

A continuous random variable: $X(\theta) = \theta$



Discrete Random Variables



A Discrete Random Variable

A discrete random variable X is a function $X(s): \Omega \rightarrow \mathbb{R}$, maps an outcome from a finite or countably infinite sample space to a number on the real line $X(s) \in \mathbb{R}$

Toss a coin 3 times in sequence

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

➤ $X(s)$ = the number of Heads in the sequence

$$X(HTH) = 2 \quad X(THT) = 1 \quad \dots$$



Discrete Random Variables



A Discrete Random Variable

A discrete random variable X is a function $X(s): \Omega \rightarrow \mathbb{R}$, maps an outcome from a finite or countably infinite sample space to a number on the real line $X(s) \in \mathbb{R}$

Toss a coin 3 times in sequence

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

➤ $X(s)$ = the number of Heads in the sequence

$$X(HTH) = 2 \quad X(THT) = 1 \quad \dots$$

➤ $X(s) \begin{cases} \text{The index of the first H} \\ 0 \text{ if the sequence has no H} \end{cases}$

$$X(TTH) = 3 \quad X(TTT) = 0 \quad \dots$$



Discrete Random Variables



A Discrete Random Variable

A **discrete random variable** X is a function $X(s): \Omega \rightarrow \mathbb{R}$, maps an outcome from a finite or countably infinite sample space to a number on the real line $X(s) \in \mathbb{R}$

Toss a coin 3 times in sequence

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

➤ $X(s)$ = the number of Heads in the sequence

$$X(s) = 2$$

$\Rightarrow$ corresponds to the event $s = \{HHT, HTH, THH\}$

$$1 < X(s) \leq 3$$

$\Rightarrow s = \{HHH, HHT, HTH, THH\}$



Discrete Random Variables



A Discrete Random Variable

Probability measure on random variables

Toss a coin 3 times in sequence

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

➤ $X(s)$ = the number of Heads in the sequence

$X(s) = 2 \Rightarrow$ corresponds to the event $s = \{HHT, HTH, THH\}$

$$\Rightarrow P(X=2) = P(\{HHT, HTH, THH\}) = 3/8$$

$1 < X(s) \leq 3 \Rightarrow s = \{HHH, HHT, HTH, THH\}$

$$\Rightarrow P(1 < X \leq 3) = P(\{HHH, HHT, HTH, THH\}) = 4/8$$

Question

- $P(X=1)$
- $P(X=2)$
- $P(X < 3)$
- $P(X \leq -1)$
- $P(X \leq 3)$
- $P(1 < X \leq 3)$



Probability Mass Function

Probability mass function

$$p_X(x) = P(X = x)$$

Toss a coin 3 times in sequence

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$X(s)$ = the number of Heads in the sequence

- $p(0) = 1/8$
- $p(1) = 3/8$
- $p(2) = 3/8$
- $p(3) = 1/8$



Probability Distribution Function

Probability mass function

$$p_X(x) = P(X = x)$$

(Cumulative) Probability distribution function

$$F_X(x) = P(X \leq x)$$

Toss a coin 3 times in sequence

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$X(s)$ = the number of Heads in the sequence

➤ $p(0) = 1/8$

➤ $p(1) = 3/8$

➤ $p(2) = 3/8$

➤ $p(3) = 1/8$

➤ $F(-1) = P(X \leq -1) = 0/8$

➤ $F(0) = P(X \leq 0) = 1/8$

➤ $F(1) = P(X \leq 1) = 4/8$

➤ $F(2) = P(X \leq 2) = 7/8$

➤ $F(3) = P(X \leq 3) = 1$

➤ $F(4) = P(X \leq 4) = 1$



Probability Distribution Function

(Cumulative) probability distribution function

$$F_X(x) = P(X \leq x)$$

Toss a coin 3 times in sequence

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$X(s)$ = the number of Heads in the sequence

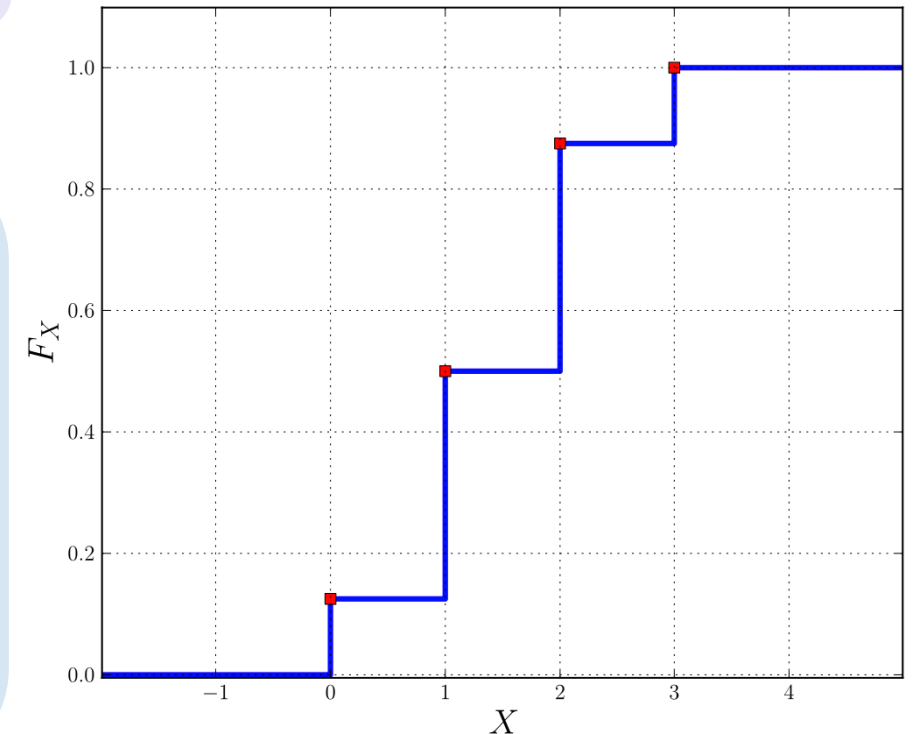
$$\text{➤ } F(-1) = P(X \leq -1) = 0/8 \quad \text{➤ } F(2) = P(X \leq 2) = 7/8$$

$$\text{➤ } F(0) = P(X \leq 0) = 1/8 \quad \text{➤ } F(3) = P(X \leq 3) = 1$$

$$\text{➤ } F(1) = P(X \leq 1) = 4/8 \quad \text{➤ } F(4) = P(X \leq 4) = 1$$

$$P(0 < X \leq 2) = P(X = 1) + P(X = 2) = F(2) - F(0)$$

The graph of the probability distribution function



Discrete Random Variables



Practice

Throw a die twice. X is sum of two numbers. $P[X=7]$?

Throw a die twice in sequence

$$\Omega = \{(1,1), (1,2), \dots, (2,1), (2,2), \dots, (6,1), (6,2), \dots, (6,6)\} \quad \Rightarrow \quad n(\Omega) = 6 \cdot 6 = 36$$

X = sum of two numbers

$$P[X=7] = P[\{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}]$$

$$P[X=7] = P[(1,6)] + P[(2,5)] + P[(3,4)] + P[(4,3)] + P[(5,2)] + P[(6,1)]$$

$$P[X = 7] = \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} = \frac{1}{6}$$



Practice

Throw a die twice. X is sum of two numbers. $F[3]$?

Throw a die twice in sequence

$$\Omega = \{(1,1), (1,2), \dots, (2,1), (2,2), \dots, (6,1), (6,2), \dots, (6,6)\} \quad \Rightarrow \quad n(\Omega) = 6 * 6 = 36$$

X = sum of two numbers

$$F(3) = P[X \leq 3] = P[2] + P[3] = P[(1,1)] + P[\{(1,2), (2,1)\}]$$

$$F(3) = P[X \leq 3] = P[(1,1)] + P[(1,2)] + P[(2,1)]$$

$$F(3) = P[X \leq 3] = \frac{1}{36} + \frac{1}{36} + \frac{1}{36} = \frac{1}{12}$$



Practice

We extract 3 balls from a box containing an equal number of white and black balls, and we put the ball back into the box after each extraction. How many white balls can occur and what are the associated probabilities?

X = Number of white balls

Sample space	X	Pro	Sample space	X	Pro	X	0	1	2	3
WWW	3	1/8	WBB	1	1/8	P(X)	1/8	3/8	3/8	1/8
WWB	2	1/8	BWB	1	1/8					
WBW	2	1/8	BBW	1	1/8					
BWW	2	1/8	BBB	0	1/8					

Outline

SECTION 1

Discrete Random Variables

SECTION 2

Mathematical Expectations

SECTION 3

Covariance & Correlation



Mathematical Expectations



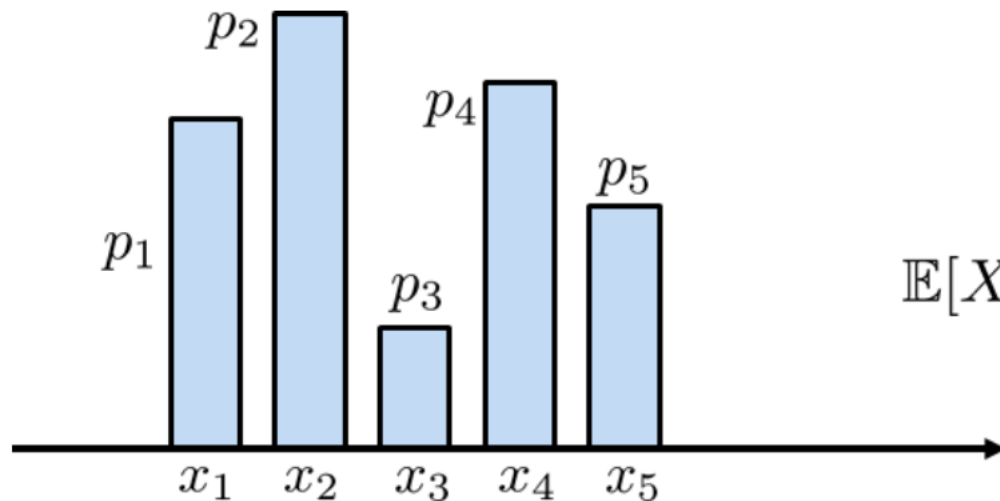
Expectation

The expected value of a discrete random variable X is

$$E[X] = \sum_k x_k \cdot P(X = x_k) = \sum_k x_k \cdot p_X(x_k)$$

The $E[X]$ represents the weighted average value of X

$E[X]$ is also called the mean of X



$$\mathbb{E}[X] = p_1 x_1 + \dots + p_5 x_5$$

Mathematical Expectations



Expectation

The expected value of a discrete random variable X is

$$E[X] = \sum_k x_k \cdot P(X = x_k) = \sum_k x_k \cdot p_X(x_k)$$

Rolling a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

x	1	2	3	4	5	6
p(x)	1/6	1/6	1/6	1/6	1/6	1/6

$$E[X] = 1 * \frac{1}{6} + 2 * \frac{1}{6} + 3 * \frac{1}{6} + 4 * \frac{1}{6} + 5 * \frac{1}{6} + 6 * \frac{1}{6} = \frac{7}{2}$$

Prove the following:

- $E[\alpha X] = \alpha E[X]$
- $E[\alpha X + b] = \alpha E[X] + b$

Mathematical Expectations



Expectation

The expected value of a discrete random variable X is

$$E[X] = \sum_k x_k \cdot P(X = x_k) = \sum_k x_k \cdot p_X(x_k)$$

Toss a coin 3 times in sequence

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$X(s)$ = the number of Heads in the sequence

- $p(0) = 1/8$
- $p(1) = 3/8$
- $p(2) = 3/8$
- $p(3) = 1/8$

x	0	1	2	3
p(x)	1/8	3/8	3/8	1/8

$$E[X] = 0 * \frac{1}{8} + 1 * \frac{3}{8} + 2 * \frac{3}{8} + 3 * \frac{1}{8} = \frac{3}{2}$$

Mathematical Expectations



Variance

Mean of X:

$$\mu = E[X]$$

Variance (The average weighted **square distance** from the mean) of X:

$$\text{Var}(X) = E[(X - \mu)^2] = \sum_k (x_k - \mu)^2 p(x_k)$$

Toss a coin 2 times in sequence

$$\Omega = \{HH, TH, HT, TT\}$$

X(s) = the number of Heads in the sequence

➤ $p(0) = 1/4$

➤ $p(1) = 2/4$

➤ $p(2) = 1/4$

x	0	1	2
p(x)	1/4	2/4	1/4

$$E[X] = 0 * \frac{1}{4} + 1 * \frac{2}{4} + 2 * \frac{1}{4} = 1$$

$$\text{Var}(X) = (0 - 1)^2 * \frac{1}{4} + (1 - 1)^2 * \frac{2}{4} + (2 - 1)^2 * \frac{1}{4} = 0.5$$

Mathematical Expectations



Variance

Mean of X:

$$\mu = E[X]$$

Variance (The average weighted **square distance** from the mean) of X:

$$\text{Var}(X) = E[(X - \mu)^2] = \sum_k (x_k - \mu)^2 p(x_k) = E[X^2] - \mu^2$$

Toss a coin 2 times in sequence

$$\Omega = \{HH, TH, HT, TT\}$$

$X(s)$ = the number of Heads in the sequence

- $p(0) = 1/4$
- $p(1) = 2/4$
- $p(2) = 1/4$

x	0	1	2
p(x)	1/4	2/4	1/4

$$E[X] = 0 * \frac{1}{4} + 1 * \frac{2}{4} + 2 * \frac{1}{4} = 1$$

$$E[X^2] = 0 * \frac{1}{4} + 1 * \frac{2}{4} + 4 * \frac{1}{4} = 1.5 \quad \text{Var}(X) = 1.5 - 1 = 0.5$$



Standard Deviation

Standard Deviation (The average weighted **distance** from the mean) of X:

$$\sigma(X) = \sqrt{\text{Var}(X)} = \sqrt{E[(X - \mu)^2]} = \sqrt{E[X^2] - \mu^2}$$

Toss a coin 2 times in sequence

$$\Omega = \{HH, TH, HT, TT\}$$

X(s) = the number of Heads in the sequence

- $p(0) = 1/4$
- $p(1) = 2/4$
- $p(2) = 1/4$

x	0	1	2
p(x)	1/4	2/4	1/4

$$E[X] = 0 * \frac{1}{4} + 1 * \frac{2}{4} + 2 * \frac{1}{4} = 1$$

$$\text{Var}(X) = 1.5 - 1 = 0.5$$

$$\sigma(X) = \sqrt{1.5 - 1} = \sqrt{0.5} = 0.707$$



Standard Deviation

Standard Deviation (The average weighted **distance** from the mean) of X:

$$\sigma(X) = \sqrt{Var(X)} = \sqrt{E[(X - \mu)^2]} = \sqrt{E[X^2] - \mu^2}$$

Rolling a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

x	1	2	3	4	5	6
p(x)	1/6	1/6	1/6	1/6	1/6	1/6

$$E[X] = \frac{7}{2}$$

$$E[X^2] = \sum_{k=1}^6 \left[k^2 * \frac{1}{6} \right] = \frac{(1 + 4 + 9 + 16 + 25 + 36)}{6} = \frac{91}{6}$$

Mathematical Expectations



Standard Deviation

Standard Deviation (The average weighted **distance** from the mean) of X:

$$\sigma(X) = \sqrt{\text{Var}(X)} = \sqrt{E[(X - \mu)^2]} = \sqrt{E[X^2] - \mu^2}$$

Rolling a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

x	1	2	3	4	5	6
p(x)	1/6	1/6	1/6	1/6	1/6	1/6

$$E[X] = \frac{7}{2}$$

$$E[X^2] = \frac{91}{6}$$

$$\text{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$$

$$\sigma(X) = \sqrt{\frac{35}{12}} \approx 1.7$$

Mathematical Expectations



Standard Deviation

$$\mathbf{X} = \{1, 3, 4, 4\}$$

$$E[X] = \mu = \sum_k x_k \cdot p_X(x_k) \quad \text{Var}(X) = \sum_k (x_k - \mu)^2 p(x_k) \quad \sigma(X) = \sqrt{\text{Var}(X)}$$

x	1	3	4
times	1	1	2
p(x)	1/4	1/4	1/2

$$E[X] = \mu = 1 * \frac{1}{4} + 3 * \frac{1}{4} + 4 * \frac{1}{2} = 3$$

$$\begin{aligned} \text{Var}(X) &= \frac{1}{4} * (1 - 3)^2 + \frac{1}{4} * (3 - 3)^2 \\ &\quad + \frac{1}{2} * (4 - 3)^2 = 1.5 \end{aligned}$$

$$\sigma(X) = \sqrt{\text{Var}(X)} = \sqrt{1.5} \approx 1.22$$

Mathematical Expectations



Standard Deviation

$$\mathbf{X} = \{1, 3, 4, 4\}$$

$$E[X] = \mu = \sum_k x_k \cdot p_X(x_k) \quad \text{Var}(X) = \sum_k (x_k - \mu)^2 p(x_k) \quad \sigma(X) = \sqrt{\text{Var}(X)}$$

$$\text{Mean: } \mu = \frac{1}{n} \sum_k x_k$$

$$\text{Variance: } \text{Var}(X) = \frac{1}{n} \sum_k (x_k - \mu)^2$$

$$\text{Standard Deviation: } \sigma(X) = \sqrt{\text{Var}(X)}$$

$$\mu = \frac{1}{4} (1 + 3 + 4 + 4) = 3$$

$$\begin{aligned} \text{Var}(X) &= \frac{1}{4} [(1 - 3)^2 + (3 - 3)^2 \\ &\quad + (4 - 3)^2 + (4 - 3)^2] = 1.5 \end{aligned}$$

$$\sigma(X) = \sqrt{\text{Var}(X)} = \sqrt{1.5} \approx 1.22$$

Mathematical Expectations



Standard Deviation

$$\mathbf{X} = \{1, 3, 4, 4\}$$

$$E[X] = \mu = \sum_k x_k \cdot p_X(x_k) \quad \text{Var}(X) = \sum_k (x_k - \mu)^2 p(x_k) \quad \sigma(X) = \sqrt{\text{Var}(X)}$$

$$\text{Mean: } \mu = \frac{1}{n} \sum_k x_k$$

$$\text{Variance: } \text{Var}(X) = \frac{1}{n} \sum_k (x_k - \mu)^2$$

$$\text{Standard Deviation: } \sigma(X) = \sqrt{\text{Var}(X)}$$

```
1 data = np.array([1, 3, 4, 4])
2 print(data)
3
4 print("Mean: ", np.mean(data))
5 print("Std: ", np.std(data))
6 print("Variance: ", np.var(data))
```

```
[1 3 4 4]
Mean: 3.0
Std: 1.224744871391589
Variance: 1.5
```

Mathematical Expectations



Practice

$$\mathbf{X} = \{1, 2, 8, 5\}$$

$$E[X] = \mu = \sum_k x_k \cdot p_X(x_k) \quad \text{Var}(X) = \sum_k (x_k - \mu)^2 p(x_k) \quad \sigma(X) = \sqrt{\text{Var}(X)}$$

$$\text{Mean: } \mu = \frac{1}{n} \sum_k x_k$$

$$\text{Variance: } \text{Var}(X) = \frac{1}{n} \sum_k (x_k - \mu)^2$$

$$\text{Standard Deviation: } \sigma(X) = \sqrt{\text{Var}(X)}$$

$$\mu = \frac{1}{4} (1 + 2 + 8 + 5) = 4$$

$$\text{Var}(X) = \frac{1}{4} [(1 - 4)^2 + (2 - 4)^2 + (8 - 4)^2 + (5 - 4)^2] = 7.5$$

$$\sigma(X) = \sqrt{\text{Var}(X)} = \sqrt{7.5} \approx 2.74$$

Mathematical Expectations



Practice

Find the variance and standard deviation of the following probability distribution table?

X	0	1	2	3
P(X)	0.1	0.2	0.4	0.3

$$E[X] = \mu = \sum_k x_k \cdot p_X(x_k)$$

$$\text{Var}(X) = \sum_k (x_k - \mu)^2 p(x_k)$$

$$\sigma(X) = \sqrt{\text{Var}(X)}$$

$$\begin{aligned} E[X] \\ &= 0 * 0.1 + 1 * 0.2 + 2 * 0.4 + 3 * 0.3 = 1.9 \end{aligned}$$

$$\begin{aligned} \text{Var}(X) \\ &= (0 - 1.9)^2 * 0.1 + (1 - 1.9)^2 * 0.2 \\ &\quad + (2 - 1.9)^2 * 0.4 + (3 - 1.9)^2 * 0.3 = 0.89 \end{aligned}$$

$$\sigma(X) = \sqrt{0.89} = 0.94$$

Mathematical Expectations



Practice

Flip a coin with probability p to get a head. Let X be a random variable denoting the outcome.

Find $E[X]$ and $\text{Var}[X]$ based on the PMF of X is

$$p_X(0) = 1 - p, \quad p_X(1) = p$$

$$E[X] = \mu = \sum_k x_k \cdot p_X(x_k)$$

$$\text{Var}(X) = E[X^2] - \mu^2$$

$$\sigma(X) = \sqrt{\text{Var}(X)}$$

$$\begin{aligned} E[X] &= 0 * p_X(0) + 1 * p_X(1) \\ &= 0 * (1 - p) + 1 * p = p \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E[X^2] - E[X]^2 \\ &= 0^2 * p_X(0) + 1^2 * p_X(1) - E[X]^2 \\ &= 0 * (1 - p) + 1 * p - p^2 = p(1 - p) \end{aligned}$$

QUIZ TIME

Outline

SECTION 1

Discrete Random Variables

SECTION 2

Mathematical Expectations

SECTION 3

Covariance & Correlation



Covariance



Correlation

Covariance & Correlation



Covariance

X, Y: random variables

$$E[X] = \mu_X; E[Y] = \mu_Y$$

Covariance of X and Y:

Sample

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = \frac{\sum (x - \mu_X)(y - \mu_Y)}{n - 1}$$

$$X = \{1, 3, 4, 4\}$$

$$E[X] = \mu = \frac{1}{4}(1 + 3 + 4 + 4) = 3$$

$$\text{Var}(X) = \frac{1}{4}[(1 - 3)^2 + (3 - 3)^2 + (4 - 3)^2 + (4 - 3)^2] = 1.5$$

$$\sigma(X) = \sqrt{\text{Var}(X)} = \sqrt{1.5} \approx 1.22$$

$$Y = \{1, 2, 3, 2\}$$

$$E[Y] = \mu = \frac{1}{4}(1 + 2 + 3 + 2) = 2$$

$$\text{Var}(Y) = \frac{1}{4}[(1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2 + (2 - 2)^2] = 0.5$$

$$\sigma(Y) = \sqrt{\text{Var}(Y)} = \sqrt{0.5} \approx 0.707$$

Covariance & Correlation



Covariance

X, Y: random variables

$$E[X] = \mu_X; E[Y] = \mu_Y$$

$$\mathbf{X} = \{1, 3, 4, 4\}$$

$$E[X] = \mu_X = 3$$

$$\mathbf{Y} = \{1, 2, 3, 2\}$$

$$E[Y] = \mu_Y = 2$$

Covariance of X and Y:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = \frac{\sum (x - \mu_X)(y - \mu_Y)}{n - 1}$$

$$\text{Cov}(X, Y) = \frac{\sum (x - \mu_X)(y - \mu_Y)}{n - 1}$$

$$\begin{aligned} &= \frac{(1 - 3)(1 - 2) + (3 - 3)(2 - 2) \\ &\quad + (4 - 3)(3 - 2) + (4 - 3)(2 - 2)}{4 - 1} \\ &= \frac{2 + 1}{3} = 1 \end{aligned}$$

Covariance & Correlation

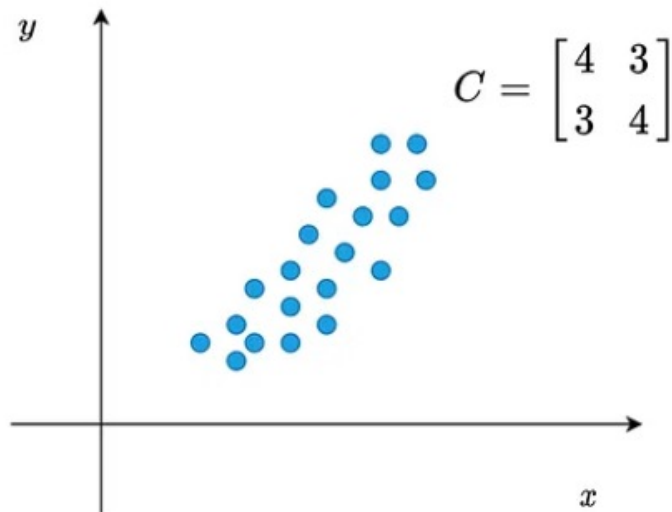


Covariance

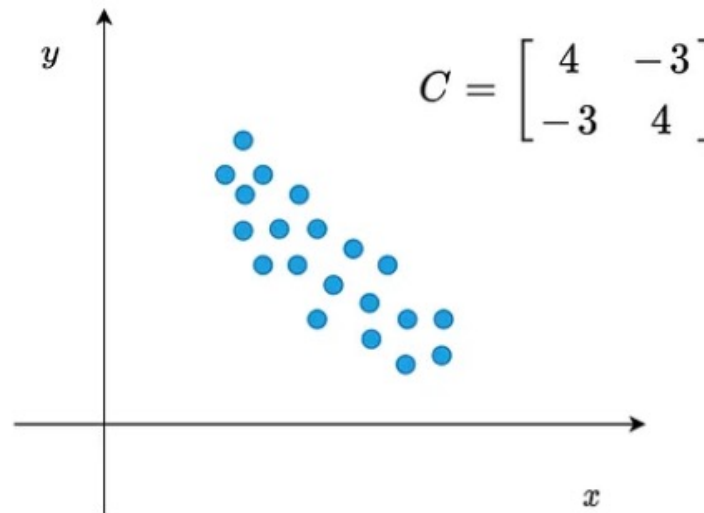
$\text{Cov}(X, Y)$
measures “concordance” or
“coherence” of X and Y

Covariance of X and Y :

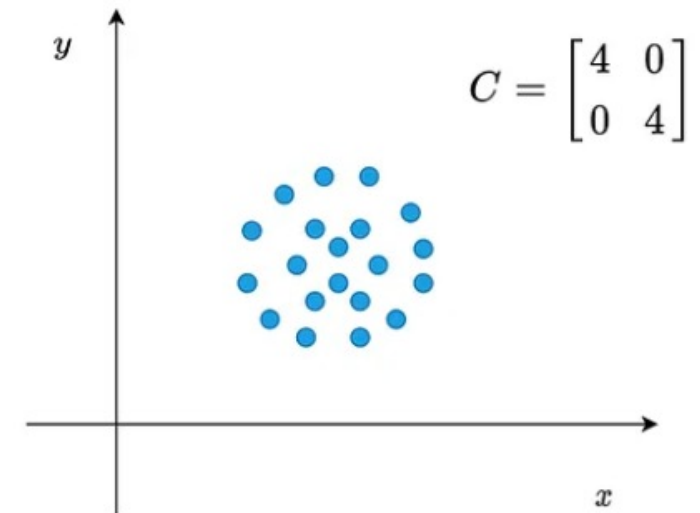
$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = \frac{\sum (x - \mu_X)(y - \mu_Y)}{n - 1}$$



Positive Covariance – $\text{Cov}(X, Y) > 0$



Negative Covariance – $\text{Cov}(X, Y) < 0$



Zero Covariance – $\text{Cov}(X, Y) = 0$

Covariance & Correlation



Correlation

A statistical measure that quantifies the strength and direction of a linear relationship between two random variables

$$\mathbf{X} = \{1, 3, 4, 4\}$$

$$E[X] = \mu_X = 3$$

$$\mathbf{Y} = \{1, 2, 3, 2\}$$

$$E[Y] = \mu_Y = 2$$

Correlation of X and Y:

$$\begin{aligned} \text{Corr}(X, Y) &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{var}(X)} \sqrt{\text{var}(Y)}} \\ &= \frac{n(\sum_i x_i y_i) - (\sum_i x_i)(\sum_i y_i)}{\sqrt{n \sum_i x_i^2 - (\sum_i x_i)^2} \sqrt{n \sum_i y_i^2 - (\sum_i y_i)^2}} \end{aligned}$$

$$\text{Corr}(X, Y) = \frac{27n - 96}{\sqrt{42n - 144} \sqrt{18n - 64}} = \frac{12}{\sqrt{24} \sqrt{8}} \approx 0.866$$

Covariance & Correlation

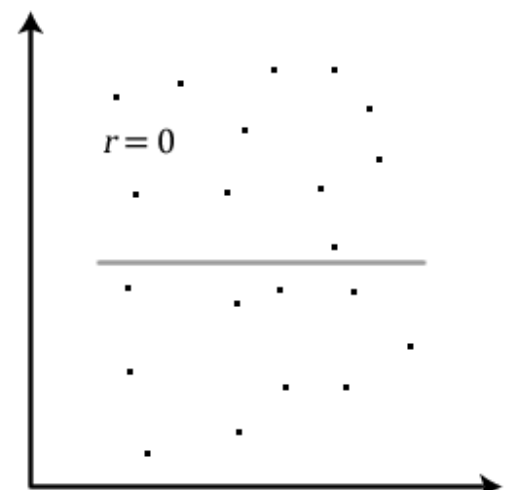
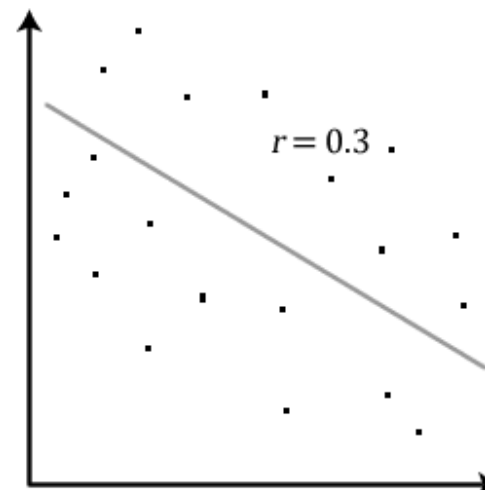
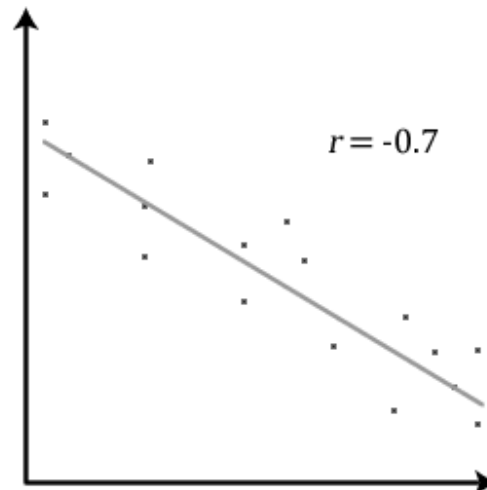
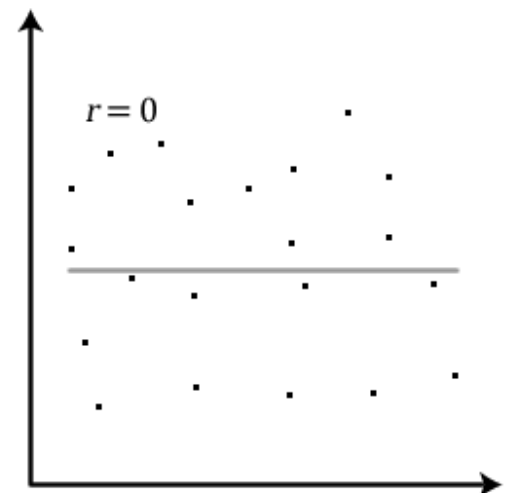
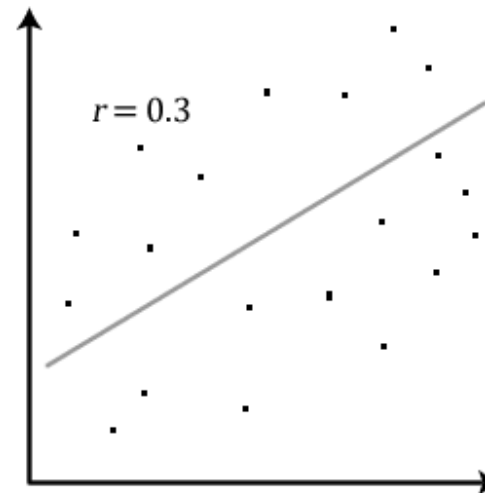
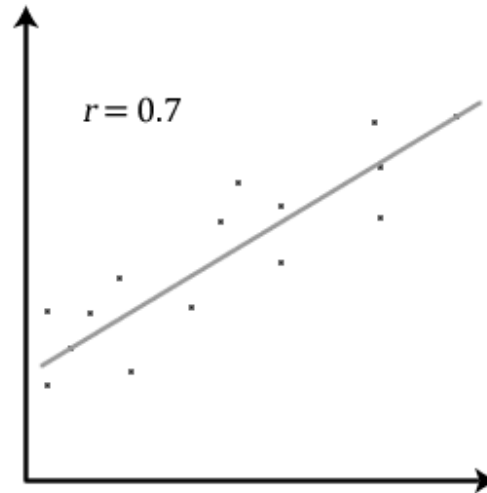


Correlation

Correlation of X and Y:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{var}(X)}\sqrt{\text{var}(Y)}}$$

$$\text{Corr}(X, Y) \in [-1, 1]$$



Covariance & Correlation



Correlation

Correlation of X and Y:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{var}(X)}\sqrt{\text{var}(Y)}}$$

$$\text{Corr}(X, Y) \in [-1, 1]$$

```
1 # Sample data for two variables
2 var1 = np.array([1, 3, 4, 4])
3 var2 = np.array([1, 2, 3, 2])
4
5 # Compute covariance
6 print("Covariance: \n", np.cov(var1, var2))
7
8 # Compute correlation coefficient
9 print("Correlation: \n", np.corrcoef(var1, var2))
```

Covariance:

```
[[2.         1.         ]
 [1.         0.66666667]]
```

Correlation:

```
[[1.         0.8660254]
 [0.8660254  1.         ]]
```

Covariance & Correlation



Practice

$$X = \{1, 1, 2, 4\}$$

$$Y = \{2, 3, 4, 3\}$$

$$E[X] = \mu_X = 2$$

$$E[Y] = \mu_Y = 3$$

$$\text{Cov}(X, Y) = \frac{\sum (x - \mu_X)(y - \mu_Y)}{n - 1}$$

$$= \frac{(1 - 2)(2 - 3) + (1 - 2)(3 - 3) + (2 - 2)(4 - 3) + (4 - 2)(3 - 3)}{4 - 1}$$

$$= \frac{1}{3}$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$= \frac{n(\sum_i x_i y_i) - (\sum_i x_i)(\sum_i y_i)}{\sqrt{n \sum_i x_i^2 - (\sum_i x_i)^2} \sqrt{n \sum_i y_i^2 - (\sum_i y_i)^2}}$$

$$= \frac{25n - 96}{\sqrt{22n - 64} \sqrt{38n - 144}} = \frac{4}{\sqrt{24}\sqrt{8}}$$

$$\approx 0.288$$

Summary

Introduction

- ❖ Random Variable
- ❖ Discrete Random Variable
- ❖ Continuous Random Variable
- ❖ Probability Mass Function
- ❖ Probability Distribution Function
- ❖ Mean

$$\mu = \frac{1}{n} \sum_k x_k$$

- ❖ Variance

$$\text{Var}(X) = \frac{1}{n} \sum_k (x_k - \mu)^2$$

- ❖ Standard Deviation

$$\sigma(X) = \sqrt{\text{Var}(X)}$$

Application

- ❖ Covariance & Correlation

$$\begin{aligned} \text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= \frac{\sum (x - \mu_X)(y - \mu_Y)}{n - 1} \end{aligned}$$

$$\begin{aligned} \text{Corr}(X, Y) &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{var}(X)} \sqrt{\text{var}(Y)}} \\ &= \frac{n(\sum_i x_i y_i) - (\sum_i x_i)(\sum_i y_i)}{\sqrt{n \sum_i x_i^2 - (\sum_i x_i)^2} \sqrt{n \sum_i y_i^2 - (\sum_i y_i)^2}} \end{aligned}$$



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Thanks!

Any questions?